

Chapter: Unified Payments Interface (UPI) Architecture

A Comprehensive Technical and Functional Architectural Breakdown

1. Introduction

The Unified Payments Interface (UPI) is an open-loop, real-time, single-window mobile payment system developed by the National Payments Corporation of India (NPCI) and regulated by the Reserve Bank of India (RBI). It merges multiple banking services, seamless fund routing, and merchant payments into a single mobile application.

Architecturally, UPI functions as an overlay network built on top of the Immediate Payment Service (IMPS) core infrastructure. Unlike traditional payment rails (like cards or wire transfers) that rely on heavy batch processing or siloed communication links, UPI leverages a secure, API-driven, distributed architecture. It decouples the user identity from actual banking credentials by utilizing a Virtual Payment Address (VPA), or UPI ID (e.g., username@bank), ensuring high security, abstracting complex settlement layers, and rendering zero-friction instant interbank transfers.

2. Customer Journey

To understand the structural orchestration of UPI, we trace the end-to-end journey of a customer performing a standard Person-to-Merchant (P2M) or Person-to-Person (P2P) push transaction via a QR code scan:

- 1 Initiation:** The customer opens their chosen UPI application (e.g., Google Pay) and triggers the camera scanner to read a merchant's static or dynamic Quick Response (QR) code.
- 2 Parsing & Data Extraction:** The application parses the payload embedded within the QR code. This payload contains critical strings: the merchant's VPA, legal name, Merchant Category Code (MCC), transaction reference ID, and optionally, the pre-defined transaction amount.
- 3 Amount Entry & Authentication:** If a static QR was scanned, the user enters the desired transaction amount. The user then reviews the transaction details on the application screen and confirms the intent to pay by initiating the authorization process.
- 4 UPI PIN Validation:** The application invokes a secure, isolated Common Library SDK provided by NPCI. The user inputs their 4 or 6-digit cryptographic UPI PIN. This PIN is instantly encrypted asymmetric-key wise inside the runtime memory before leaving the user device, preventing the host application from capturing it.

- 5 Processing & Notification:** The transaction is routed through the ecosystem participants. Within 2 to 4 seconds, the customer receives a real-time push notification indicating success, accompanied by a unique 12-digit Retrieval Reference Number (RRN).

3. Google Pay Role (TPAP)

Google Pay operates as a Third-Party Application Provider (TPAP). In the UPI ecosystem topology, a TPAP does not hook up directly to the central NPCI switch, nor does it hold settlement accounts or handle raw customer funds. Instead, it serves as the consumer-facing touchpoint and orchestrates user experience.

The primary responsibilities of Google Pay include:

- **Onboarding & Identity Management:** Managing user device registration, binding device hardware tokens to SIM cards, and linking verified phone numbers to bank accounts via cellular SMS handshakes.
- **VPA Lifecycle Allocation:** Partnering with multiple Sponsor Banks (PSP banks) to assign users a customized UPI ID extension (e.g., user@oksbi or user@okaxis) under a Multi-Bank model to balance operational load.
- **UI/UX Representation:** Rendering the transaction interfaces, parsing incoming deep links and QR specifications, and handling customer relationship management (CRM) for disputed charges.

4. NPCI Role (The Central Switch)

The National Payments Corporation of India (NPCI) acts as the central clearinghouse, message router, and settlement manager. It owns and operates the core UPI central software switch. All inter-entity communications must transit through NPCI's highly scalable network nodes.

Key architectural components managed by NPCI include:

- **Central Mapper (Address Resolution Engine):** A highly optimized, distributed internal database that maps a user's VPA directly to their structural banking coordinates (Account Number, IFSC Code, and Mobile Number). When a payment request arrives, NPCI queries this mapper to identify the precise recipient bank.
- **Request Orchestration and Routing:** Acting as a central message broker, converting, validating, and forwarding ISO 20022 compliant XML message envelopes between PSPs and core banking networks.
- **Clearing and Net Settlement:** Aggregating payment volumes across all participating banks throughout the business day, computing net debit/credit positions, and posting finalized settlement requests to the Reserve Bank of India's RTGS system for interbank fund migration.

5. Issuer Bank Role (Remitter Bank)

The Issuer Bank (or Remitter Bank) is the financial institution holding the customer's funded savings or current account. It acts as the ultimate gatekeeper of the source capital.

Its responsibilities within a transaction block are:

- **Core Banking System (CBS) Integration:** Executing real-time ledger checks. When a debit instruction arrives via its UPI payment gateway, it validates whether the customer account holds sufficient balances or credit extensions.
- **Cryptographic PIN Verification:** Hosting the Hardware Security Module (HSM) that decrypts the secure payload sent by the NPCI Common Library SDK and confirms the validity of the entered UPI PIN.
- **Atomic Debit Processing:** Executing an instant, non-reversible debit from the account balance and transmitting a synchronous debit success confirmation message back to the central NPCI switch.

6. Acquirer Bank Role (Beneficiary PSP Bank)

The Acquirer Bank (or Beneficiary Bank) is the financial entity representing the merchant or the receiving party. It operates the server-side payment infrastructure that accepts the incoming capital stream.

Its specific technical roles involve:

- **Merchant Onboarding and KYC:** Verification of merchant business documentation, provisioning merchant accounts, and registering the merchant profiles onto the central NPCI directory.
- **Credit Orchestration:** Processing incoming success instructions from the NPCI switch, identifying the target merchant ledger via its internal maps, and performing a real-time credit entry into the merchant's deposit account.
- **Instant Callback Notifications:** Pushing low-latency webhook messages to the merchant's server endpoints or transaction aggregators to announce immediate payment resolution.

7. Merchant Role

The Merchant sits at the termination end of a P2M transaction loop, consuming the payment status to authorize the release of goods or digital services. Merchants interface with UPI using specialized collection endpoints:

- **Static QR Codes:** Low-cost, non-powered visual displays containing fixed string vectors (VPA and Name). The customer determines the price point at runtime.
- **Dynamic QR Codes:** Dynamically computed, software-generated visual vectors containing a pre-populated transaction amount and unique order IDs. This enables automated reconciliation since the incoming payment maps exactly to one open invoice.
- **Intent/SDK Integration:** For mobile applications (e.g., e-commerce apps), merchants call a localized Android/iOS intent, passing payment payloads directly to on-device TPAP apps, eliminating manual QR scans.

8. Authorization Process (Step-by-Step API Flow)

The following sequence explains the structural execution flow across the system nodes during a real-time push transaction. Every message packet is mapped using structural APIs structured on standard XML schemas handled over secured HTTPS Mutual TLS tunnels.

System Architecture Flow Protocol

1. **GPay to Remitter PSP:** Customer confirms payment. GPay sends an API payload (ReqPay) to its Sponsor PSP Bank containing the encrypted UPI PIN, Remitter VPA, and Beneficiary VPA.
2. **Remitter PSP to NPCI Switch:** The Sponsor Bank signs and forwards the normalized transaction block to the central NPCI switch.
3. **NPCI Address Resolution:** NPCI validates the message signature, checks its internal Central Mapper database, resolves the Beneficiary VPA, and identifies the target Acquirer Bank.
4. **NPCI to Issuer Bank (Debit Request):** NPCI forwards an isolated debit instruction packet (ReqTxnPin) directly to the Remitter/Issuer Bank's UPI Gateway.
5. **Issuer Bank Verification:** The Issuer Gateway handles HSM processing to verify the PIN. The Core Banking System (CBS) locks the funds and executes an atomic ledger debit. It replies synchronously with a RespPay (Success) to NPCI.
6. **NPCI to Acquirer Bank (Credit Request):** Upon registering the debit success, NPCI shifts the transaction state and dispatches a credit execution instruction (ReqCredit) to the Beneficiary/Acquirer Bank.
7. **Acquirer Bank Ledger Update:** The Acquirer Bank processes the inbound packet, credits the merchant's underlying account via its internal systems, and replies with a success confirmation string back to NPCI.
8. **Final Resolution Loop:** NPCI broadcasts the finalized transaction resolution packet to both the initiating Remitter PSP (which notifies Google Pay via webhook) and the Acquirer Bank (which alerts the merchant terminal).

9. Fraud Checkpoints

Due to the instant, irreversible nature of UPI, defensive risk checks are distributed across multiple processing junctions to intercept malicious actors before ledger execution:

A. Device-to-SIM Fingerprinting (TPAP and PSP Level)

During installation, the TPAP application extracts hardware unique identifiers (IMEI, Android ID, UUID) and forces a silent background outbound SMS from the device SIM. The telecommunication network validates that the physical SIM sending the text matches the phone number registered on the account profile. If a user

attempts to log into an account from a remote device without possession of the physical SIM, authentication fails instantly.

B. Velocity and Value Threshold Controls (NPCI and Bank Level)

Automated algorithms continuously monitor transaction patterns against mathematical velocity ceilings. Limits are dynamically enforced on specific operational windows:

$$\text{Total Daily Transacted Value} \leq \text{INR } 100,000$$

$$\text{Transaction Count per 24 hr Window} \leq 10$$

Additional defensive constraints restrict newly registered profiles or SIM card migrations to an initial threshold of INR 5,000 during the first 24 hours to mitigate SIM-swap vulnerabilities.

C. VPA Risk Scoring and Blacklists (NPCI Central Engine)

Before forwarding any transaction request to the issuer, NPCI's central risk management framework computes a fraud probability score based on structural metadata. If the beneficiary VPA has been flagged for abnormal peer-to-peer collection loops or has high chargeback reporting rates, the transaction is rejected at the switch layer.

10. Product Manager Perspective

From a Product Management standpoint, building a UPI infrastructure requires balancing conversion metrics against security protocols. A Product Manager evaluates the ecosystem using performance KPIs:

Key Performance Indicator (KPI)	Target Metric Threshold	Product Management Focus Areas
Technical Decline (TD) Rate	< 1.5%	Isolating core banking timeouts, API infrastructure latency, and connection drop-offs. Managing multi-bank auto-routing networks.
Business Decline (BD) Rate	Monitor Closely	Analyzing user-driven errors such as insufficient balances, incorrect UPI PIN entries, or breach of velocity limits. Improves UI/UX alerts.
Transaction Success Rate (TSR)	> 95.5% (P2M)	The ultimate metric for customer trust. Maximized by predicting bank downtime and gracefully degrading or steering users away from failing rails.
Mean Time to Resolution (MTTR)	< 500ms (Core Processing)	Streamlining SDK footprints, optimizing network calls, and reducing payload overhead to ensure lightning-fast processing speeds.

Product Managers must engineer resilience patterns such as *Sponsor Bank Circuit Breakers*. If Bank A's system APIs degrade, the system dynamically switches traffic allocation to Bank B's active virtual pipelines, preventing platform-wide failure spikes.

11. Fraud Analyst Perspective

A Fraud Analyst views the UPI grid as an adversarial landscape where immediate payment resolution creates a high-velocity environment for illicit fund routing. Analysts track specific threat profiles:

- **Mule Account Ring Operations:** Fraudsters purchase or exploit unmonitored bank accounts from low-income demographics to receive stolen funds, quickly layering the money across dozens of sub-VPAs within minutes. Analysts flag accounts exhibiting sudden high-volume P2P velocity after long periods of dormancy.
- **Collect-Request Exploitation:** Social engineering attacks where bad actors initiate a ReqPay (Collect Request) to a victim, masquerading it as an "incentive credit" or "lottery payout." The victim mistakenly inputs their UPI PIN, which authorizes an immediate debit instead of an anticipated credit.
- **Reverse-Engineering Mapping:** Malicious actors executing continuous lookup scripts (VPA validation queries) against the NPCI switch to extract valid user account names mapped to phone databases for targeted phishing.

Analysts deploy continuous behavioral profiling tools, checking anomalies in user typing rhythms, screen layout interaction dynamics, geographical IP jumps against cellular tower locations, and correlation of device IDs to many disparate bank profiles.

12. Interview Questions and Sample Answers

Q1: What is the core difference between a TPAP and a PSP Bank in the UPI framework?

Answer: A Third-Party Application Provider (TPAP) is a customer-facing digital layer (like Google Pay) that handles user acquisition and interfaces but does not directly connect to the core central banking settlement network. A Payment Service Provider (PSP) Bank is a banking entity that possesses a banking license and hooks directly into the central NPCI switch. The TPAP must route all its transactions through its partner PSP Bank's infrastructure nodes via secured APIs.

Q2: How does UPI ensure that a user's UPI PIN cannot be intercepted or stolen by the TPAP application?

Answer: UPI enforces strict structural separation by utilizing a standardized Common Library SDK provided directly by NPCI, which is embedded inside the TPAP app but isolated at runtime. When a user enters their UPI PIN, it is processed entirely inside this secure library wrapper and immediately encrypted using an asymmetric public key provided by the Issuer Bank's HSM. The TPAP app container only sees a heavily encrypted hexadecimal data payload, making intercepting or logging the raw PIN impossible.

Q3: Explain what a Technical Decline (TD) is and how you would architect a system to minimize its impact on a merchant checkout platform.

Answer: A Technical Decline occurs when a transaction fails due to infrastructural errors independent of the customer's intent, such as internal server timeouts, core banking down-times, network switches dropping packets, or integration errors. To minimize TD impacts, a merchant checkout platform should implement: 1) A Multi-Bank Acquirer Model to dynamically route transactions away from degraded payment switches. 2) Real-time health monitoring systems that probe bank API latencies and auto-toggle active routing configurations. 3) Smart retry logic for transient network failures that checks transaction status before prompting the user to retry, preventing double debits.

13. Key Takeaways

- **Decoupled Identity Model:** By utilizing the Virtual Payment Address (VPA), UPI successfully separates user access presentation from the underlying core banking infrastructure, enhancing safety and usability.
- **Atomic 2-Step Settlement:** Transactions are processed via low-latency, real-time message switching using ISO-compliant protocols, performing real-time account updates followed by deferred multilateral bank clearing.
- **Distributed Risk Architecture:** Device-to-SIM binding, dynamic velocity rules, asymmetric cryptographic layers, and predictive machine learning models safeguard transactions across all endpoints.
- **Open Ecosystem Design:** The structural separation of TPAPs, PSP Banks, and the central NPCI infrastructure enables a competitive fintech ecosystem that maintains strict compliance, performance, and scale.